Vinitha Gadiraju

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EDUCATION

University of Colorado, Boulder (CU Boulder) August 2018 - Present
Ph.D. Computer Science – Advised by Shaun Kane
Expected Graduation Date: May 2023

University of Oregon September 2015 – June 2018
B.S. Computer Science, Minor in Psychology

PUBLICATIONS

Conference Papers

[C.3] Vinitha Gadiraju, Jérémie Garcia, Shaun K. Kane, Anke M. Brock. 2021. "It is fascinating to make these beasts fly": Understanding Visually Impaired People's Motivations and Needs for Drone Piloting. In Proceedings of the 2021 SIGACCESS Conference on Computers and Accessibility (ASSETS '21). ACM. DOI: https://doi.org/10.1145/3441852.3471219

[C.2] Vinitha Gadiraju, Olwyn Doyle, and Shaun K. Kane. 2021. Exploring Technology Design for Students with Vision Impairment in the Classroom and Remotely. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). ACM. DOI: https://doi.org/10.1145/3411764.3445755

[C.1] Vinitha Gadiraju, Annika Muehlbradt, and Shaun K. Kane. 2020. BrailleBlocks: Computational Braille Toys for Collaborative Learning. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20). ACM. DOI: https://doi.org/10.1145/3313831.3376295

Posters, Extended Abstracts, and Presentations

[P.3] Vinitha Gadiraju. 2021. Tactile Play and Literacy Learning for Visually Impaired Children. Doctoral Consortium at the 23rd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '21).

[P.2] Vinitha Gadiraju. 2019. BrailleBlocks: Braille Toys for Cross-Ability Collaboration. In Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19), (pp. 688-690). ACM. DOI: https://doi.org/10.1145/3308561.3356104

[P.1] Vinitha Gadiraju and Shaun K. Kane. 2019. Computational Tools to Enable Learning Braille. Poster at CRA-W Grad Cohort for Women Cohort.

RESEARCH

Graduate Researcher, Superhuman Computing Lab

University of Colorado, Boulder, Fall 2018 - Current

- Currently exploring literacy and Expanded Core Curriculum education for blind or visually impaired children that come from bilingual/immigrant backgrounds in the United States

    - Interviewing, observing, and conducting design probes with students, their families, and educators.
- Conducted an ethnography study at the Colorado School for the Deaf and the Blind [C.2]
    - Explored learning, teaching, and collaboration strategies between peers and educators related to Braille education, literacy, and Expanded Core Curriculum
    - Collected artifacts from the classroom and conducted informal interviews with students, educators, and school administration
    - Co-designed with educators to create games and activities to teach literacy and social skills for remote education and in-person instruction
- Developed BrailleBlocks, a learning tool using tangible blocks and games to help visually impaired children and sighted collaborators learn, teach, and practice Braille [C.1]
    - Built tangible blocks representing Braille cells and an interface with educational games
    - Programmed with OpenCV to color detect pegs in the blocks for real-time Braille letter and word detection
    - Conducted user studies with sighted parents and visually impaired children

Chateaubriand Research Fellow

ENAC Toulouse, France, Spring 2021 – Summer 2021

- Explored accessible drone piloting for people with vision impairment
- Surveyed and interviewed visually impaired participants to understand their motivation to fly and how piloting can be made accessible [C.3]
- Prototyped an accessible drone piloting system with audio and haptic feedback
    - Conducted participatory design based user studies with visually impaired adults

Undergraduate Research Assistant, Human-Computer Interaction Lab

University of Oregon, Spring 2017 - Fall 2017

- Built 3 working media device prototypes for people with Rett Syndrome
- Analyzed and coded video data to study the effectiveness of individualized devices for participants

Undergraduate Research Assistant, Learning Lab

University of Oregon, Fall 2016 - Spring 2018

- Coded in ELAN to analyze audio files of an infant's surrounding for music
- Conducted independent project in the lab studying infant exposure to happy, upbeat music in their daily lives. Analyzed data in R.

Undergraduate Research Assistant, Advanced Integration and Mining Laboratory

University of Oregon, Fall 2015

- Conducted literature review to identify trends between health forum/app use and health outcome

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INDUSTRY EXPERIENCE

Software Engineering Intern, Intel Corporation, Data Center Engineering Group

Hillsboro, Oregon, Summer 2016

- Created visualization software for Platform Hardware of Intel Server products.
- Drew Platform HW description graphics in browser with HTML Canvas and SVG elements developing custom JavaScript code using D3.js Library

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AWARDS

2021 **Sickle Cell Disease Challenge 1st Place Winner**, National Heart, Lung, and Blood Institute

Designed a full trivia game prototype on Figma to help patients learn about Sickle Cell Disease

2021 **Publication Recognition Award**, University of Colorado Boulder, Computer Science Department

2021 **Collegiate Award Finalist**, NCWIT

2020 **Chateaubriand Fellow**, Embassy of France in the United States

2020 **Graduate Research Fellow**, National Science Foundation

2020 **Collegiate Award Finalist**, NCWIT

2020 **Colorado Award Winner for Aspirations in Computing**, NCWIT

2019 **Graduate Student Research Competition Winner**, ACM ASSETS 2019

1st Place Winner at the ACM Graduate Student Research Competition at the ASSETS 2019 Conference. Presented a poster and gave a talk on BrailleBlocks [P.2].

TEACHING

Fall 2018 – Spring 2020	Teaching Assistant, Fundamentals of Human-Computer Interaction	CU Boulder
Spring 2017	Tutor/Manager, Fundamentals of Human-Computer Interaction	University of Oregon
Fall 2017	Tutor, Top Research Methods in Psychology	University of Oregon

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PRESS

2020 ATLAS affiliated PhD student honored by National Science Foundation

ATLAS Institute, April 15 2020

2019 BrailleBlocks: Smart Toys for Collaborative Braille Education

International Council on English Braille, December 2019 newsletter

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SERVICE

2021	Student Volunteer, ACM ASSETS	Virtual
2020	Student Volunteer, ACM CHI (Cancelled due to COVID-19)	Honolulu, USA
2020	Reviewer, ACM/IEEE HRI	Cambridge, UK

SKILLS

UI/UX design, usability studies, ethnography, interview, survey, design probe, accessibility, prototyping, Figma, OpenCV, behavioral coding, R, SPSS, Python, Java, C, C++, JavaScript, HTML, CSS

ORGANIZATIONS

Antiracism and Inclusion Committee

CU Boulder, Department of Engineering, Summer 2020 – Current

- Working with administration to analyze student survey data and faculty/course review questionnaires to identify issues with inclusivity, racism, and harassment in the department
- Advocated for and helped implement inclusive recruitment practices for students and faculty
- Designed a survey to characterize the racial climate and culture of the department

Graduate Student Advisory Board

CU Boulder, Department of Engineering, Fall 2019 – Fall 2020

- Computer Science Representative (2019-2020)

TechTogether Conference, Director

University of Oregon, Spring 2018

- Organized a keynote speaker, panels with industry professionals, 1-on-1 resume sessions, and networking events.

Women in Computer Science

University of Oregon, Fall 2016 – Spring 2018

- President (2017-2018)
- Secretary (2016-2017)

Tutoring Program Co-Creator/Team Coordinator, Psychology Department

University of Oregon, Spring 2017 - Spring 2018

- Designed a syllabus for the program, weekly presentation slides on effective tutoring strategies, trained future tutors, and held office hours.

Students of the Indian Subcontinent

University of Oregon, Fall 2017 - Spring 2018

- Vice President (2017-2018)